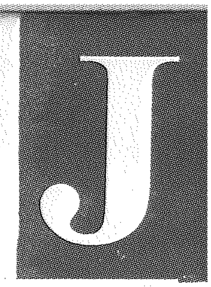

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(NMTC-Maths Olympiad at ***JUNIOR LEVEL*** *IX & X Standards)*

Saturday, 3rd September 2005.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m.- 2 hours*

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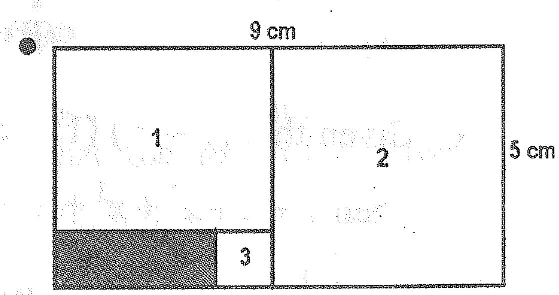
1. (22, 48), (61, 76), (29, 34) are some pairs of distinct two digit numbers whose product ends with digit 6. How many such pairs are there?

(A) 477 (B) 315 (C) 549 (D) 405

2. The digits 1,2,3,4 are used to generate 256 different 4-digit numbers. The sum of the 256 numbers in

(A) 71440 (B) 711040 (C) 704110 (D) 741040

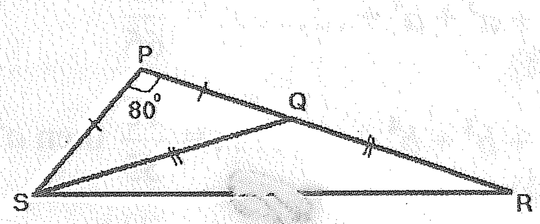
3. Find the area in square centimeters of the shaded rectangle if all other shapes 1,2,3 in the large 9cms × 5cms rectangle are squares.



(A) 4 (B) 3

(C) 2 (D) 1.5

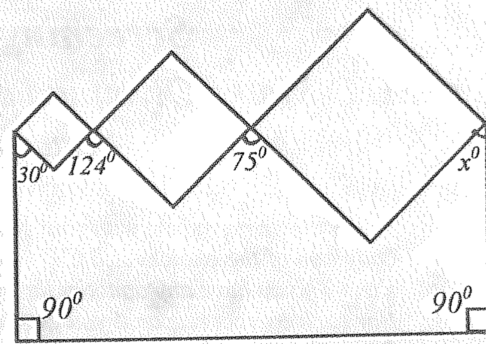
4. In the diagram $PS = PQ$ and $QS = QR$. If $\angle SPQ = 80^0$ then $\angle QRS$ equals



(A) 15^0 (B) 20^0

(C) 25^0 (D) 30^0

5. Three squares are joined together at the corners onto two vertical poles as shown aside. Find the value of x when the angles are given as shown



(A) 39 (B) 41

(C) 43 (D) 44

6. Numbers with two digits or more in which the digits reading from left to right occur in strictly increasing order are called as "sorted numbers". For example 125, 14 and 239 are sorted numbers while 255, 74 and 198 are not. Suppose that a complete list of sorted numbers in prepared and written in increasing order, the 100^{th} number on this list is

(A) 389 (B) 356 (C) 345 (D) 258

7. In how many different ways can 3 children share eight identical sweets so that each child gets at least one?

(A) 21 (B) 24 (C) 36 (D) 45

8. Given that $(1-x)(1+x+x^2+x^3+x^4) = \frac{31}{32}$ and x is a rational number.

Then $1+x+x^2+x^3+x^4+x^5$ is

(A) $\frac{31}{64}$ (B) $\frac{31}{32}$ (C) $\frac{63}{64}$ (D) $\frac{63}{32}$

9. Given

$$a+a^2+a^3+\ldots\ldots. = \frac{5}{6} \qquad a^2+a^3+a^4+\ldots\ldots. = \frac{25}{66}$$

$$b+b^2+b^3+\ldots\ldots. = \frac{6}{5} \qquad b^2+b^3+b^4+\ldots\ldots. = \frac{36}{55}$$ then a^2-b^2 is

(A) $\frac{-1}{11}$ (B) $\frac{1}{11}$ (C) $\frac{-1}{30}$ (D) $\frac{11}{30}$

10. The last two digits of $(2006)^{2005}$ is

(A) 16 (B) 36 (C) 76 (D) 96

11. A cone is made from a circular sector, by joining the two radii and the ratio of the radius and slant height of the cone is as 1:2. Then the angle of the sector from which the cone is made is

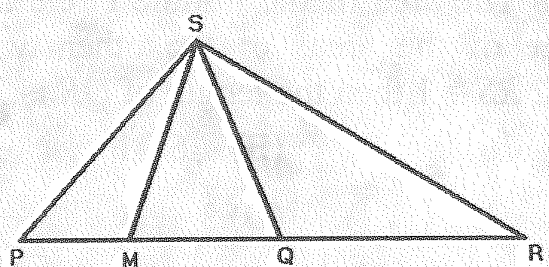
(A) 90^0 (B) 135^0 (C) 180^0 (D) 270^0

12. The number of prime numbers less than 100 which can be expressed as the sum of the squares of two natural numbers is

(A) 11 (B) 12 (C) 16 (D) 20

13. Given the equation of the circle $x^2 + y^2 = 100$, the number of points (a,b) lying on the circle, where 'a' an 'b' are both integers is

(A) 2 (B) 4 (C) 8 (D) 12

14. 

In the adjoining figure P, M, Q and R are collinear points are $PM=MQ=MS$. It is also given $SR^2 = PR.\ QR$. Then

(A) $\angle QSR = \angle MSP$ (B) $\angle QSR = \angle MSQ$

(C) $\angle QSM = \angle PSM$ (D) $\angle SQR = \angle SMP$

15. Seven consecutive numbers are chosen. From these seven numbers, two numbers are chosen. What is the probability that the sum of the two numbers chosen is divisible by 7?

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

16. If two successive discounts of a% and b% are given on a certain article, the single equivalent discount is

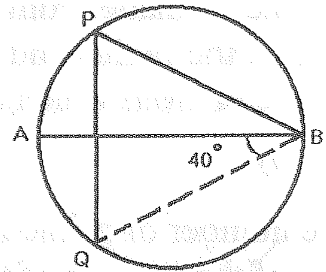
(A) $a+b+\frac{ab}{100}$ (B) $a+b-\frac{ab}{100}$

(C) $a-b+\frac{ab}{100}$ (D) $a-b-\frac{ab}{100}$

17. In a triangle ABC, $\angle A=30^0$, BC=6cm. Then the radius of the circum circle is

(A) 3cms (B) 6cms (C) 8cms (D) 4.5cms

18. AB is the diameter of the circle. Then $\angle QPB$ is



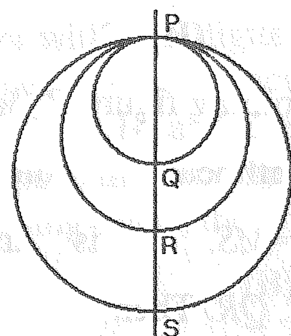
(A) 40^0 (B) 50^0

(C) 60^0 (D) 30^0

19. If $\dfrac{97}{19} = w + \dfrac{1}{x + \dfrac{1}{y}}$ where w, x, y are integers then $w + x + y$ equals

(A) 16 (B) 17 (C) 18 (D) 19

20.



$PQRS$ is a common diameter of the three circles. The area of the middle circle is the average of the areas of the other two. If $PQ = 2\ RS = 1$ then the length of QR is

(A) $1+\sqrt{6}$ (B) $\sqrt{6}-1$

(C) 4 (D) 3

21. The positive numbers x and y satisfy $xy = 1$. The minimum value of $\dfrac{1}{x^4} + \dfrac{1}{4y^4}$ is

(A) $\dfrac{1}{2}$ (B) $\dfrac{5}{8}$ (C) 1 (D) no minimum

22. Of the points (0,0), (2,0), (3,1), (1,2), (3,3), (4,3) and (2,4) at most how many can be on a circle?

(A) 5 (B) 3 (C) 4 (D) 6

23. How many of these expressions $x^3 + y^4$, $x^4 + y^3$, $x^3 + y^3$ and $x^4 - y^4$ are positive for all possible numbers x and y for which $x>y$?

(A) 1 (B) 0 (C) 2 (D) 3

24. The number of ordered pair of digits (A,B) such that $A\,3640548981270644\,B$ is divisible by 99 is

(A) 3 (B) 2 (C) 1 (D) zero

25. The roots of the equation $x^5 - 40x^4 + Px^3 + Qx^2 + Rx + S = 0$ are in geometry progression. The sum of their reciprocals is 10. Then $|S|$ is equal to

(A) 16 (B) 32 (C) 4 (D) 1